上课地点：五教5101

2023年10月11日（**周六，15:20~16:55**）

面向生物医疗应用的集成电路设计技术

王志华

zhihua@Tsinghua.edu.cn

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内容

- 什么是世界最好的大学
- 集成电路是不可替代的技术
- 历史上集成电路是怎么发展的
- 以电子技术为核心的医疗器械
- 实例：神经刺激器及智能助听
- 应用于医疗仪器的集成电路
- 医疗电子近期成果

2

什么是世界最好的大学

3

世界最好的大学的4个指标

Four Criteria of the World's Best Universities

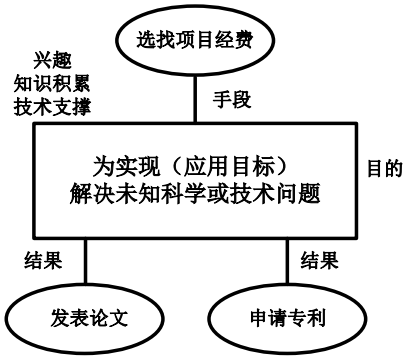
- Criterion 1
  - 全球青年學子趨之若鶩的到這個大學求學
  - Global young students rush to this university to study
- Criterion 2
  - 全球青年學者爭先恐後的到這個大學求職
  - Young scholars from all over the world are rushing to find jobs at this university.
- Criterion 3
  - 這個大學產生了很多被工業界應用的成果
  - This university has produced many results that are applied in industry
- Criterion 4
  - 这个大学的毕业生可以在世界各地找到工作，特别是学术界的工作
  - Graduates of this university can find jobs all around the world, especially in academia

在世界最好大学的工科院系怎么开展研究

关于科研目的、科研结果的公开讨论

- 【为什么申请经费】为了科研的目的，研究者需要钱和其他条件，找政府或者工业届寻求支持，从而可以开展研究。因此，申请经费是科研的『手段，而不是目的』
- 【为什么发表学术论文】科研工作取得了结果，研究者迫不及待地要让别人知道，从而也使科研成果得到初步的同行检验，这就是论文。对于参加科研工作的研究生，发表论文同时也是科研工作能力的基本训练。论文是科研的『结果之一，不是目的』。
- 【为什么申报专利】科研成果有潜在的产业价值，在公开发表论文之前申请专利，既可以保护原始研究者未来可能的商业利益，又可以让同行避免在同一研究方向耗费资源，从而有利于社会，因此专利也是科研的『结果之一，不是目的』。
- （ 2012年8月12日 @王志华_茌平）

工科研究，要把研究成果写在产品上



真正好的研究成果：两个特征

- 作为一个（目标是）世界一流大学的研究者，目标是什么？
  1. 你的研究成果，被工业界（产品）所应用
  2. 工业界（社会），承认这个成果是你的研究成果
- 怎么实现？
  1. 研究上，做世界上最好的事情
  2. 传播上让全世界知道你做的最好

Don Pederson不是David Patterson（Spice 不是 RISC-V）

王志华_茌平

2018-6-8 02:22 来自 iPhone客户端

工程学科的著名教授，其“青史留名”，不在于他发表了多少论文，更重要在于其为工业界的直接贡献。本人在2013年8月也在新浪微博上发表过“工科学者有所成”的说法。现用国外著名教授Spice之父Don Pederson教授的例子说明。 工程领域学者的成名，需对工业届有所贡献（SP...

工程领域学者的成名，需对工业届有所贡献（SPICE与D...

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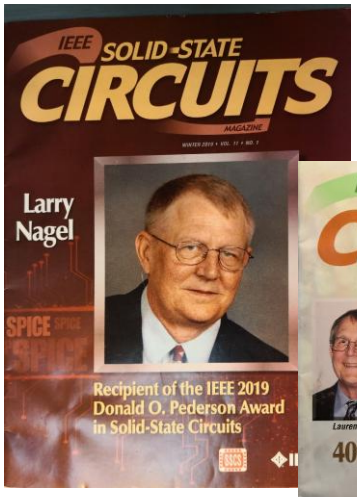
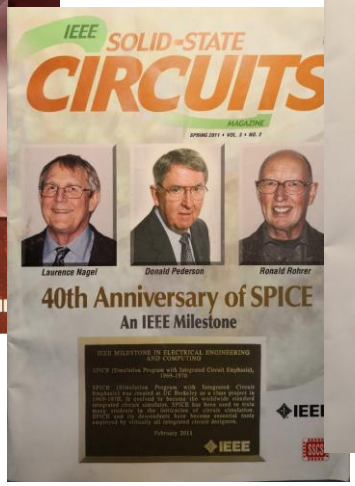
研发Spice成就的业界大咖





LAWRENCE W. NAGEL is an independent consultant in the San Francisco Bay Area. His Ph.D. dissertation involved developing the SPICE program.

WOLFRAM BLUME learned about SPICE while working at Silicon Systems in Tustin, California. Currently he pursues a variety of R&D projects at Blume Engineering in Aliso Viejo, California.

SALLY LIU is technical director of the Design Technology Division at TSMC, Hsin-Chu, Taiwan. As a graduate student, she worked on Don Pedeson's team.

TOM QUARLES develops algorithms and code for Calibre LVS at Mentor Graphics. He was the architect and primary author of SPICE3.

LYNN WEBER spent many late nights with Larry Nagel adding the noise functions into the CANCER routines. He spent most of his career with Hewlett Packard in Santa Clara and Palo Alto and with Trimble Navigation in Sunnyvale, California.

RICHARD I. DOWELL is wih Avago Technologies. He was a student at UC Berkeley, during the circuit-simulation craze that ultimately produced SPICE.

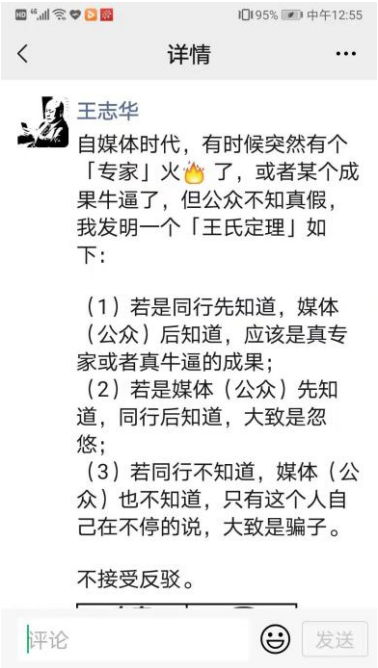
IAN GETREU runs a consulting company, IG Associates. As one of the first users of SPICE, he had the opportunity to work on the bipolar transistor model in SPICE.

KIM HAILEY is the cofounder of Meta-Software and Pixel-Cam. He started out in microprocessor design and worked at General Instrument, Xerox, Data General, and AMD.

RON ROHRER is university professor emeritus of electrical and computer engineering at Carnegie Mellon University in Pittsburgh. While a professor at UC Berkeley, he taught a course on circuit simulation.

一个好的研究成果，怎么让世界知道？

- 新“发现”的“王氏定律”
- 基本思想
  - 科学研究应该
    - 在微电子相关领域“做世界上最好的事情，同时让世界上知道我们做的最好”
  - 学术活动应该
    - 引导教授、学生们“把我们做的最好的研究成果告诉全球同行”
    - 先告诉同行→后告诉公众（反过来就可能是忽悠）
- 同行在哪里？
  - 在同行认可的期刊（会议）上发表好的论文
    - 好的学术论文的重要标志：同行评审（peer reviewed）
  - 在同行关注的学术活动上演示学术成果
    - 例如电报的发明（马可尼与波波夫）



应用需求是创新的源泉，长期坚持是成功的关键

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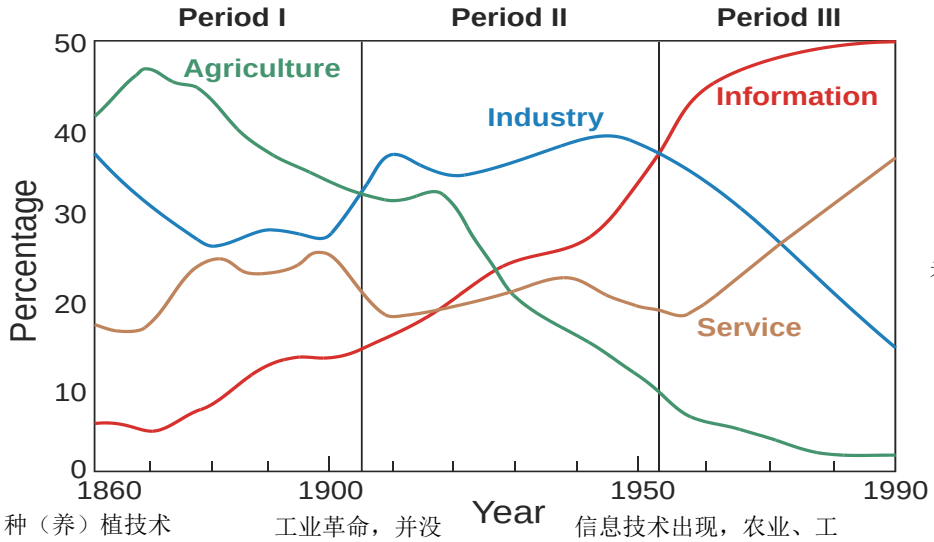
做科研的四部曲

- 阅读文献
  - 知道别人做什么，知道别人怎么做，知道世界先进水平在哪里，知道什么地方可能产生突破
- 锻炼技能
  - 工程领域的任何的创新，都是在技能提高的基础上，先明白“不创新”的东西是怎么做出来的
- 认真工作
  - 不忽视细节。（设计错误无法纠正，流片前不应存在已知错误，仿真永远是辅助，芯片测试出现错误时的仿真再现。。。）
- 成果总结
  - 论文、专利、技术报告
  - 不要赶时间发烂论文

集成电路是不可替代的技术

科学技术历史上，重要的技术发明都不可替代



未来有什么？

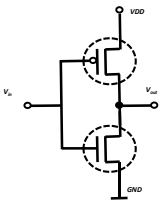
种（养）植技术带来了农耕时代

工业革命，并没有消灭农业技术

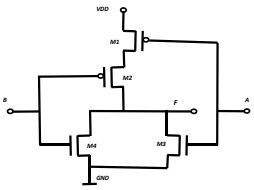
信息技术出现，农业、工业仍技术持续、稳定发展

- Changing composition of work force in the United States S.M. Sze, J. Appl. Phys. Vol. 22 (1983)

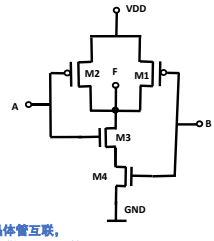
晶体管互联可以实现人类所需要的全部计算（信息获取、传输、处理、存储）



两个晶体管互联，可以实现“非not”运算

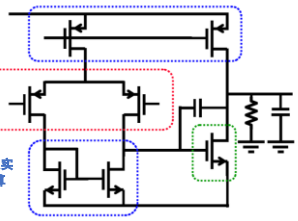


四个晶体管互联，可以实现“或or”运算



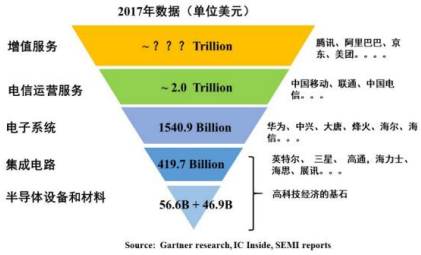
四个晶体管互联，可以实现“与and”运算

人类所需要的全部运算，都就可以用晶体管的互联实现



多个晶体管互联，可以实现“模拟analog”运算（模拟信号处理）

两个相对独立的阶段：
设计算法，解决人类需要解决的问题
制造出集成电路



集成电路支撑起信息时代。工业的基础、产业的核心、安全的保障、知识产权的有效载体，与人民生活息息相关。

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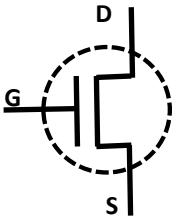
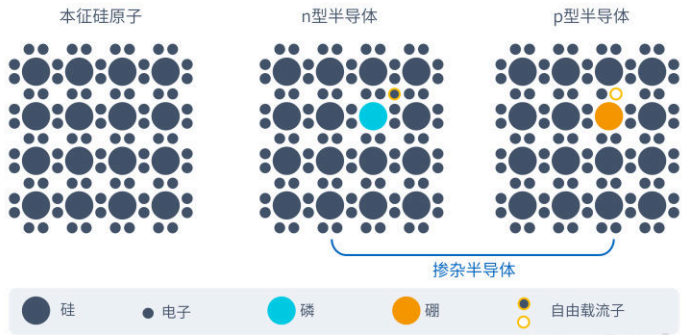
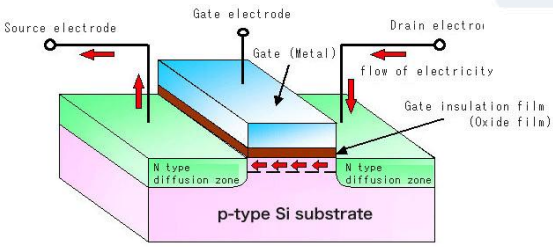


N-沟道MOS晶体管
NMOS Transistor





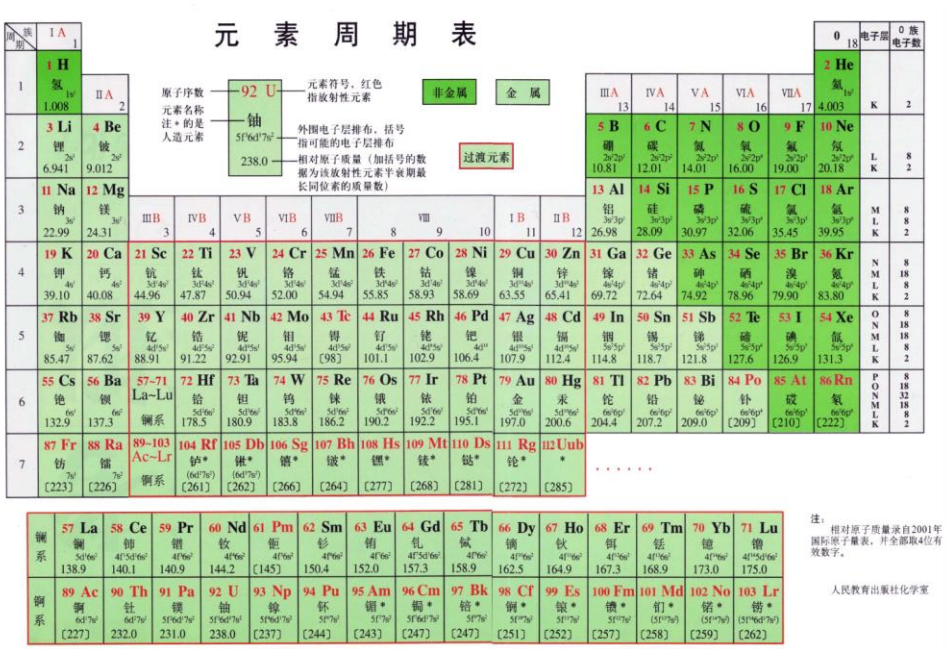
Construction of MOSFET

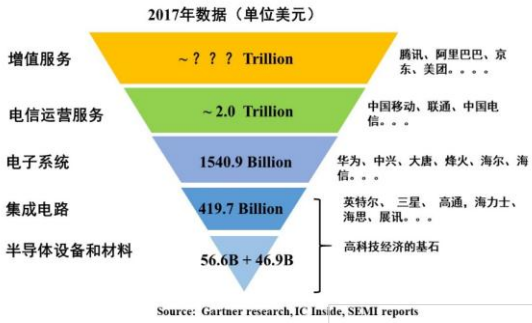
Metal-Oxide-Semiconductor Field-Effect Transistor

晶体管需要半导体

关于材料

1. 衬底材料
2. 集成电路生产过程中用到周期表中70+%的材料





智能时代的集成电路

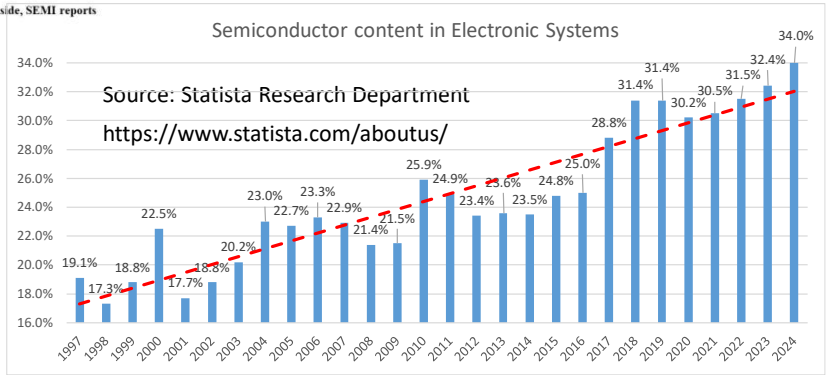
应用驱动的产品定义	算法研究	算法用晶体管互联实现	集成电路制造

向着一个万亿美元的产业逼近

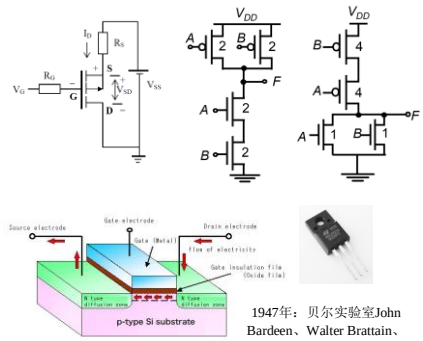
存储器进入新的发展时期

汽车半导体进入高速发展期

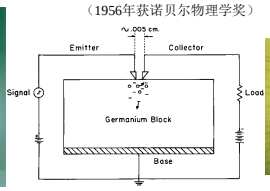
人工智能从拼算力到多样性

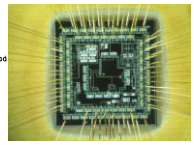


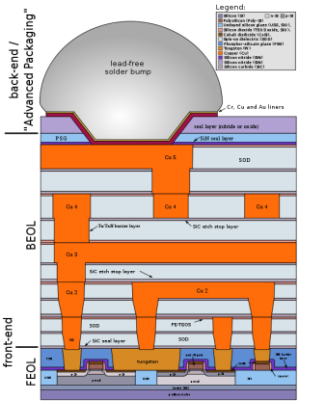
晶体管与集成电路：人类最重要的发明

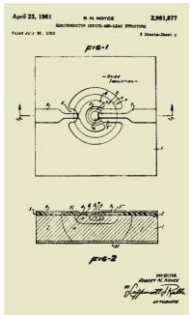


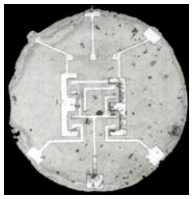
1947年：贝尔实验室John Bardeen、Walter Brattain、William Shockley等人发明了晶体管
（1956年获诺贝尔物理学奖）

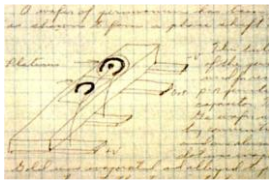


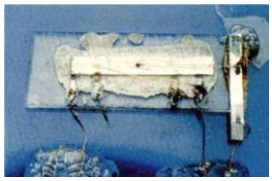












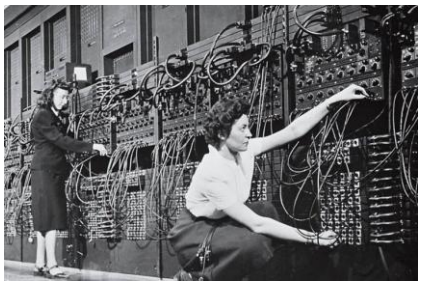
1958年：仙童公司Robert Noyce与德仪公司Jack Kilby间隔数月分别发明了集成电路。
（Jack Kilby因而获2000年诺贝尔物理学奖, Robert Noyce则因1990年已去世而失去了这一机会）

常见作物的千粒重

- 水稻 千粒重 18-34克，每公斤29400-55000粒。
- 粳稻 千粒重 25-32克，每公斤 31200-40000粒 。
- 籼稻 千粒重18-25克，每公斤 40000-55000粒 。
- 小麦 千粒重 23-58 克，每公斤 17200-43400粒。
- 玉米 千粒重 180-500 克，每公斤 2000-5560 粒 。
- 高粱 千粒重 20-34 克，每公斤 29400-50000粒 。
- 谷子 千粒重 2.2-4.0克，每公斤 250000-454000粒。
- 大麦 千粒重20-48 克，每公斤 20800-50000 粒。
- 大豆 千粒重 110-250 克，每公斤 4000-9000 粒。

1,000粒大米：	~25g
1,000,000粒大米	~25,000g(25kg)
1Billion 粒大米	~25,000公斤——？每公斤200块大洋

历史上集成电路是怎么发展的

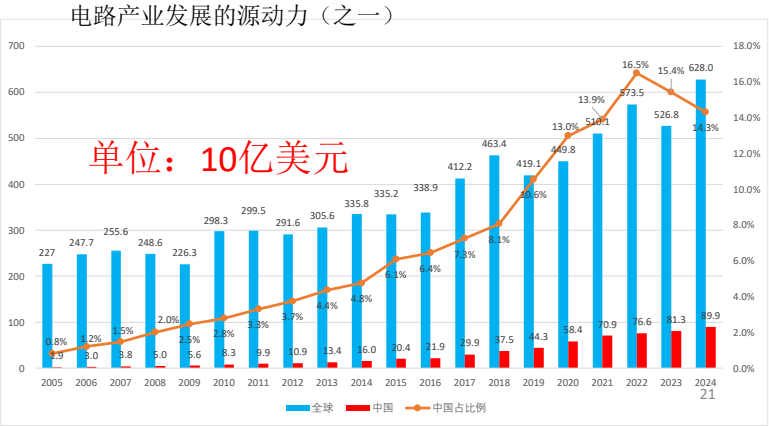


（计算设备）小型化、智能化

1946: ENIAC, using vacuum tubes, 200 bytes filling a big room and consuming 150 KW

2017: iPhone X, 3GB LPDDR4 RAM, 174 g(6.14 ounce), A11 Bionic runs on a TDP of 5W and is passively cooled.

（计算设备）小型化、智能化是（历史上）集成电路产业发展的源动力（之一）



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集成电路：计算机与互联网

信息产品	第一代	第二代	第三代	第四代	不断微型化时代	云、异构计算时代
	体积巨大 高耗能 可靠性低	体积减小 重量减轻 能耗降低 可靠性高	IBM System360 DEC PDP 系列	个人计算机(PC)革命，Apple II，IBM PC	电脑性能指数规律提升	多核、并行、异质结构
集成电路	真空管 电子管	晶体管	中小规模集成电路	微处理器与大规模集成电路	VLSI集成度增加、性能提升	后摩尔时代的异构集成（含Chiplets）
	前集成电路时代		集成电路时代			
	1940s	1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s	2030+

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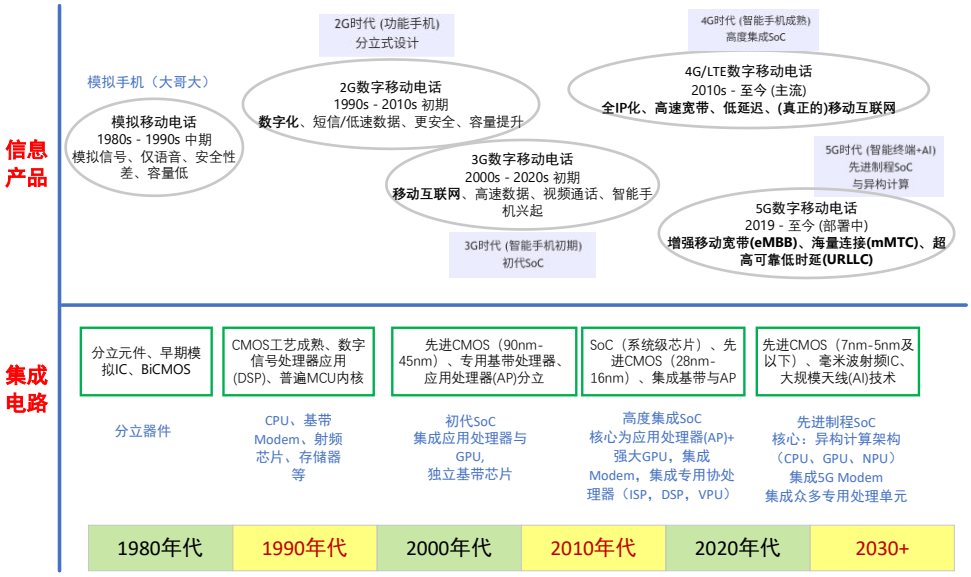
集成电路：移动通信与移动互联网



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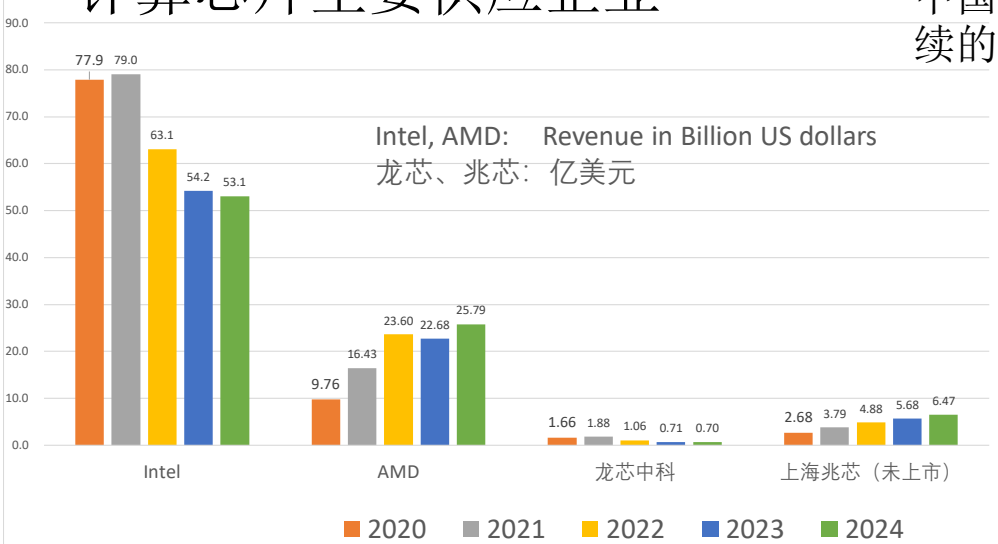
计算芯片主要供应企业

中国企业需要持续的支持和关注



左图尺度不同：

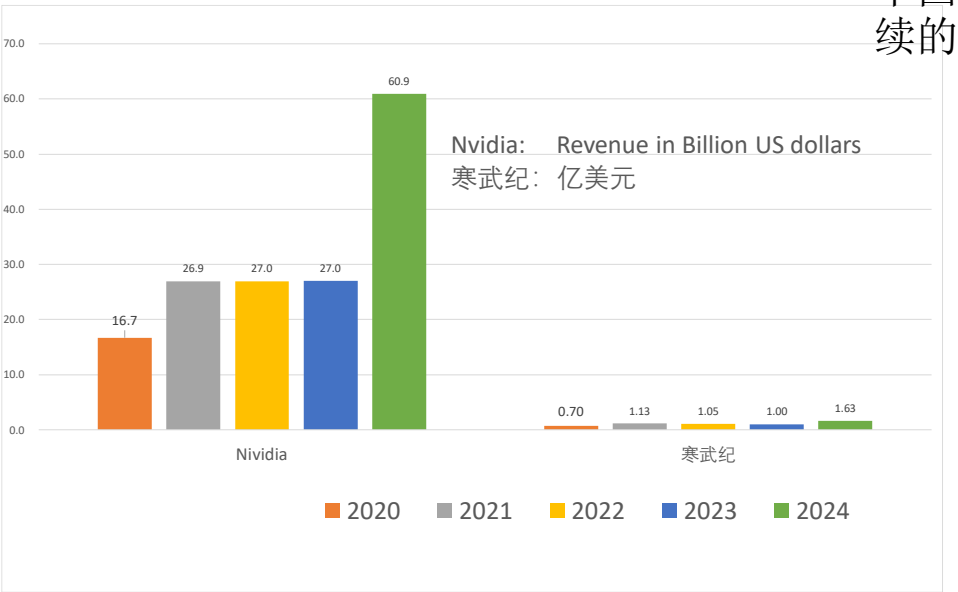
外国公司尺度10亿美元

大陆公司尺度：亿美元

24

计算芯片主要供应企业

中国企业需要持续的支持和关注

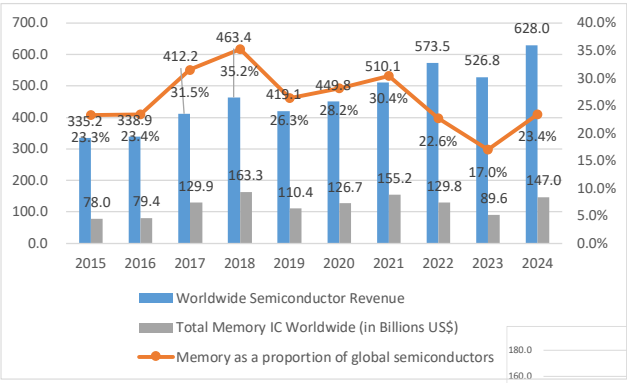


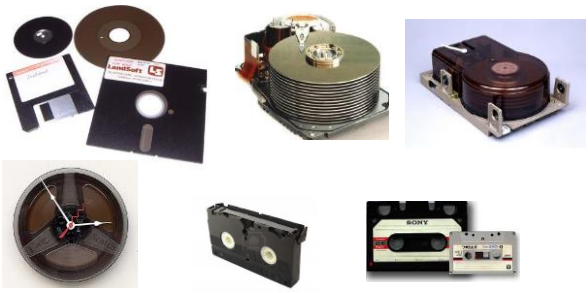
左图尺度不同：

外国公司尺度10亿美元

大陆公司尺度：亿美元

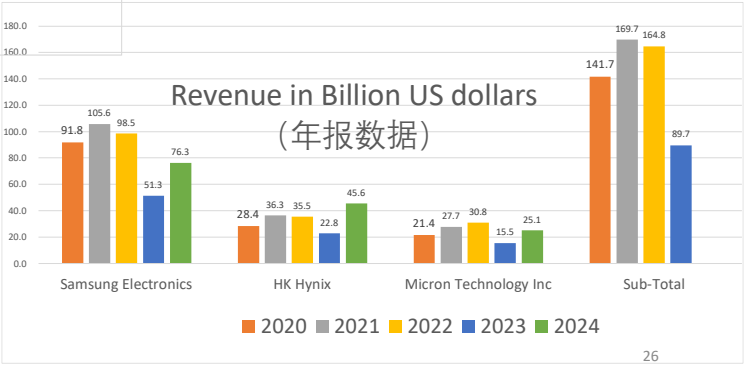
25





（非电技术）电子化：
集成电路发展的另一驱动

- 磁存储变电存储



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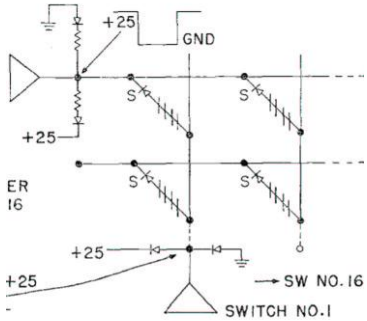
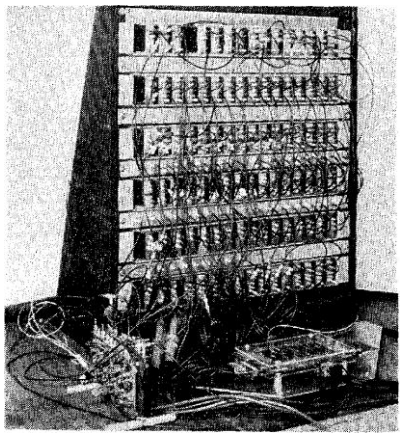
早期的电脑内存是磁环存贮器

RCA at ISSCC 1966

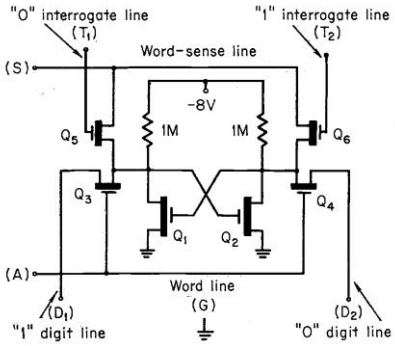


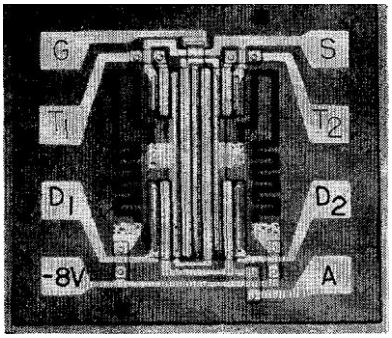


在1960年代，磁芯存储器是计算机存储单元的主流，1966年，Kaufman 和Pryor在ISSCC上宣布了64x64的新型存储器，用铁氧体做存储单元，将二极管选择阵列集成到一起。其中包含了电路、材料以及混合封装方面的创新。其效果是增加磁存储器的读写速度，减小存储器体积。上右图为256字的存储器演示系统。

“杀死”磁存储器（内存）的技术：第一个集成半导体存储单元

NEC at ISSCC 1966





采用这个电路技术，一个 4 Gb 存储器会消耗1MW的功率，占用1,000 km^2 的面积

- **电路单元类似与现代的 6T静态存储器**
  - **1mm x 0.9 mm^2 area**
  - **18 V**
  - **60 μW/bit standby power**

这是第一次发表CMOS电路的文章，此时的CMOS工艺已经出现

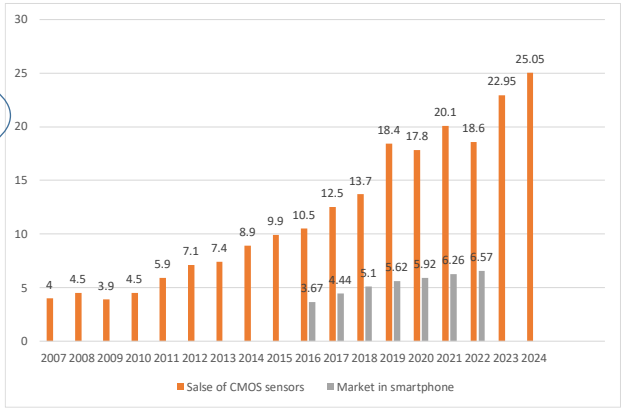
- 新功能新应用推动CMOS图像传感器产业发展，在移动设备和汽车应用的驱动下，2015年~2021年，CMOS图像（CIS）产业的复合年增长率为10.4%
- 预计市场规模将从2015年的100亿美元增长到2025年的336亿美元。

韦尔半导体（OmniVision）（虞仁荣）**2024年销售收入257.31亿元**，同比增长22.41%，全球第3

思特威（SmartSens Technology）（徐辰）**2024年销售收入59.68亿元**，同比增长108.87%

格科微电子（GalaxyCore)（赵立新）**2024年销售收入63.83亿元**，同比增长35.90%

柯达公司成立于1880年，2012年1月破产保护

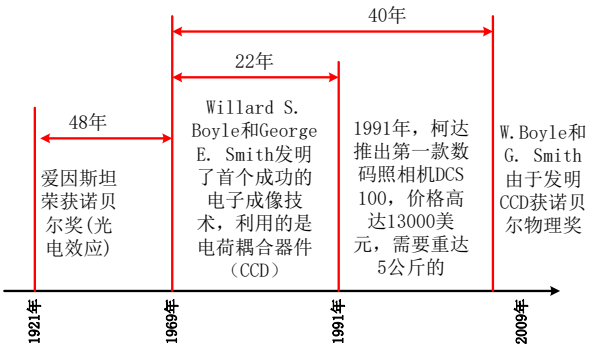


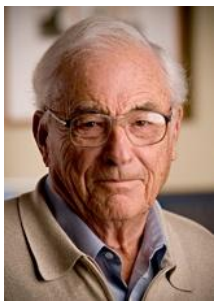
（非电技术）电子化：集成电路发展的另一驱动

- 光电变换器

“杀死”胶卷的技术：图像传感器与数码相机

- 1969年，Willard S. Boyle和George E. Smith发明了首个成功的电子成像技术，利用的是数字传感器——电荷耦合器件（CCD）。CCD技术利用了爱因斯坦荣获1921年诺贝尔奖的光电效应。
- 2009年两人获得诺贝尔物理奖



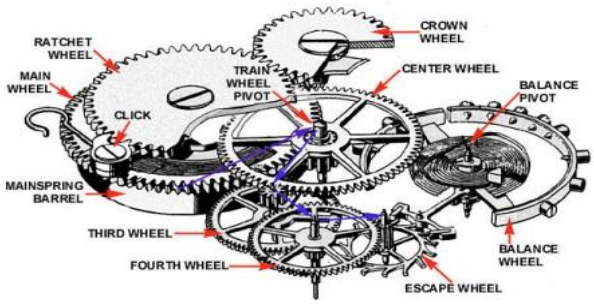


Willard S. Boyle

George E. Smith

（非电技术）电子化：集成电路发展的另一驱动

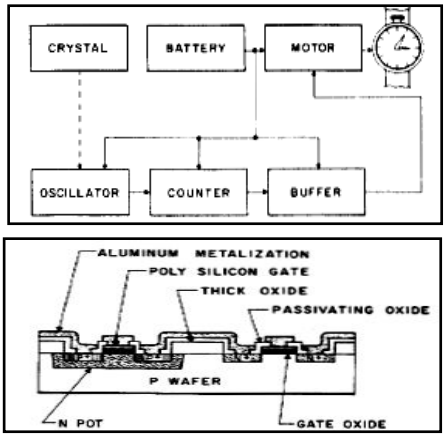
机械钟表的消失
The demise of mechanical watch

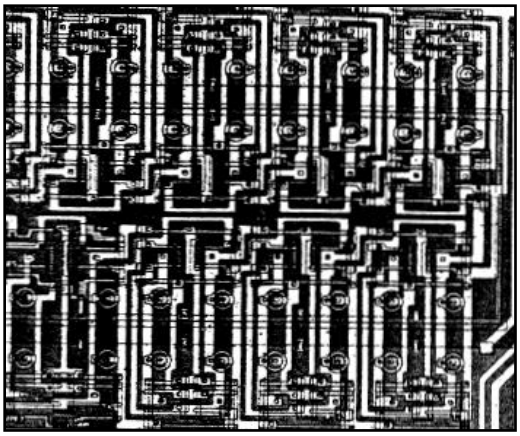


发明于15世纪的械钟表，作为计时工具，已经完全没有存在的价值

It has no any value for a mechanical watch as a timing tool

“杀死”钟表的芯片： CMOS Electronic Wristwatch
Daniels and Burgess, Motorola: 1971 ISSCC



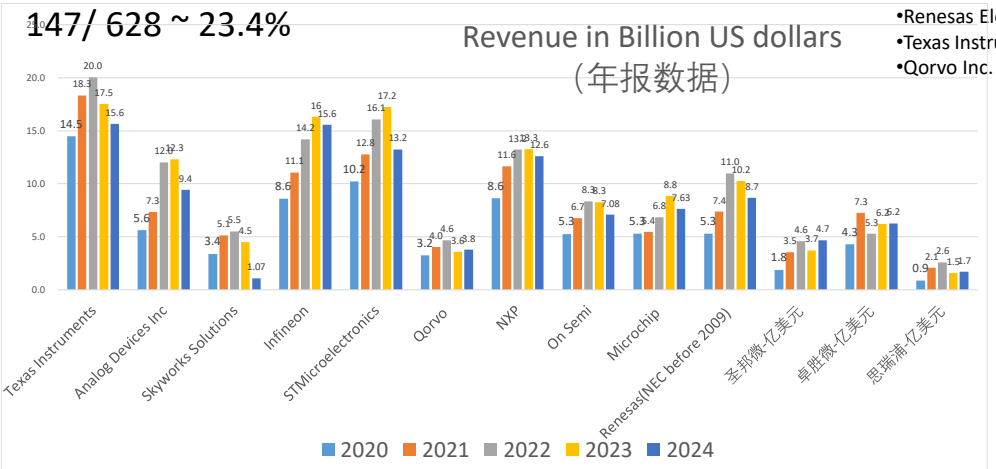


- 235 transistors, 3 mm2 die area
- 1.2 V power supply, 6 mW power dissipation

非电技术的电气化
不可缺少模拟电路

2024年10个企业的销售额之和为112 Billions US$
占集成电路销售总额23+%
147/ 628 ~ 23.4%

- Analog Devices Inc.
- Infineon Technologies AG
- Microchip Technology Inc.
- NXP Semiconductors NV
- ON Semiconductor
- Richtek Technology Corporation (MediaTek Inc.)
- Skyworks Solutions Inc.
- STMicroelectronics NV
- Renesas Electronics Corporation
- Texas Instruments Inc.
- Qorvo Inc.



左图尺度不同：

外国公司尺度10亿美元

大陆公司尺度：亿美元

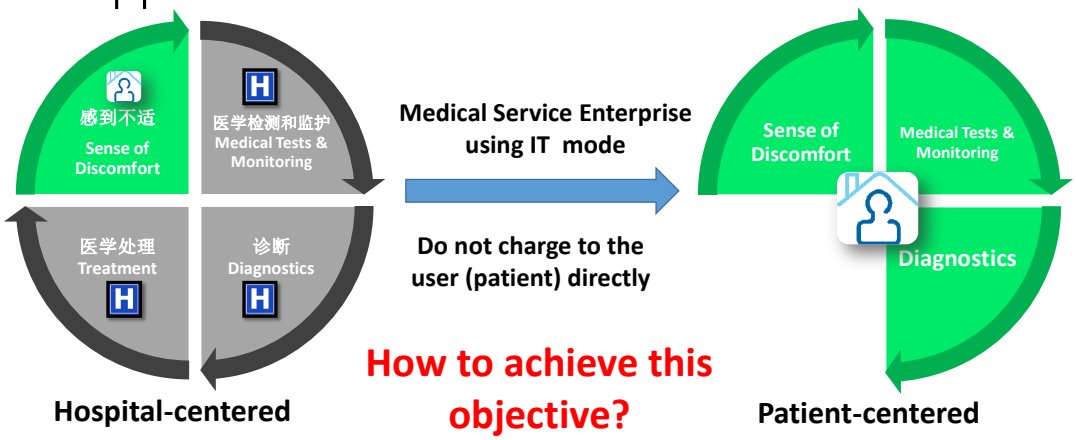
电子技术为核心的医疗器械

信息技术：信息的获取、处理、存储、传输

加个定语

生命和医疗信息的获取、处理、存储、传输

Could the model of today's information service be applied to future mobile medical services



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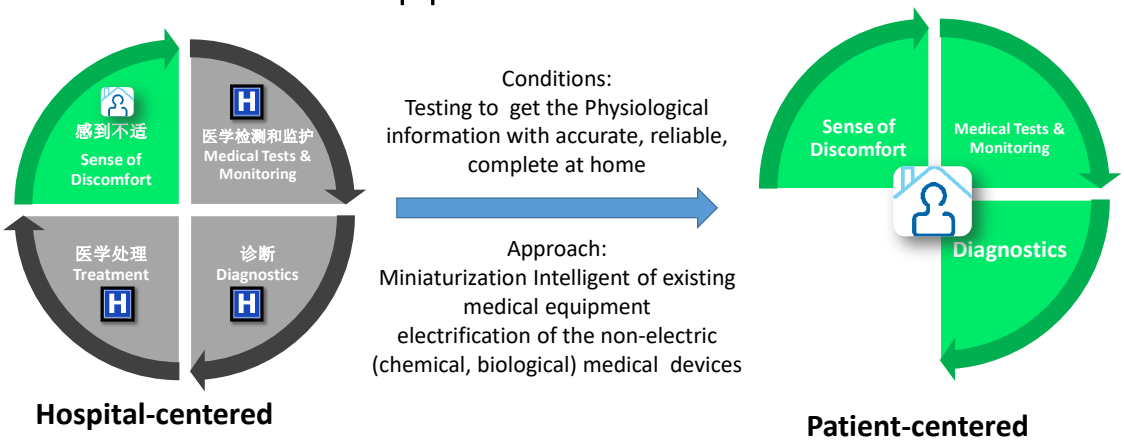
Conditions and Approach



It is not to replace today's medical services, but to provide a new type of medical service to effective use of the knowledge of medical people, and it does not directly charge to the consumers

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TECHNOLOGY ENABLERS

- Portable and/or Implantable medical monitoring systems
  - Reliable and seamless monitoring systems that can be integrated into patients daily life routine
- Ambulatory data analysis
  - Real-time data analysis and even diagnostics increasing the efficiency of health-care delivery
- Close the loop
  - Smart drug delivery and/or stimulation systems for preventive (care)

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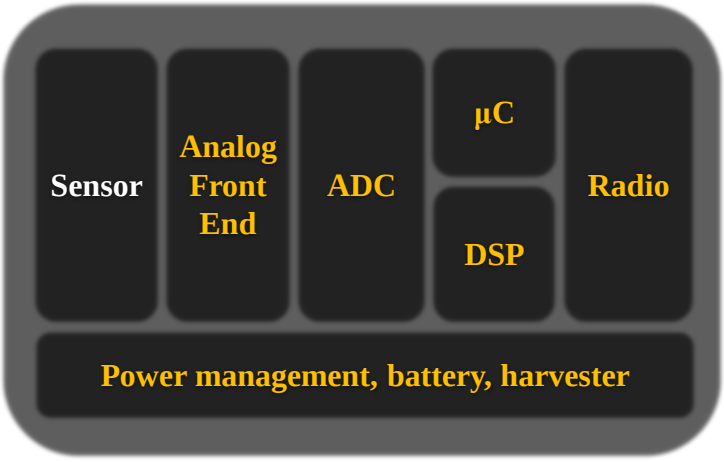
Enabling Technology is Integrated Circuit



- Information Sciences:
  Acquisition
  processing,
  Storage
  Transmission

  of (medical and life) signals

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The market size of the medical device is similar to that of the semiconductor in 2024

Rank	Companies	2024 Revenue (Billions of U.S. Dollars)	2024 Market share
1	Samsung Electronics	65.52	10.5%
2	Intel	49.19	7.9%
3	Nividia	45.99	7.3%
4	SK hynix	42.82	6.8%
5	Qualcomm	32.36	5.2%
6	Micron Technology	27.82	4.4%
7	Broadcom	27.64	4.4%
8	AMD	23.95	3.8%
9	Apple	18.88	3.0%
10	Infineon Technologies	16.00	2.6%
	Sub-Total	350.18	55.9%
	Others	275.79	44.1%
	Total Sales	**625.97**	100.0%

Rank	Companies	2024 Revenue (Billions of U.S. Dollars)	2024 Market share
1	Medtronic	33.00	5.5%
2	Johnson & Johnson	30.40	5.1%
3	Abbott Laboratories	27.90	4.7%
4	Danaher Corporation	24.00	4.0%
5	Medline Industries	23.20	3.9%
6	Stryker Corporation	23.00	3.9%
7	Siemens Healthineers	23.00	3.9%
8	Becton, Dickinson & Co	21.00	3.5%
9	GE Healthcare	19.70	3.3%
10	Philips	19.50	3.3%
	Sub-Total	244.70	41.1%
	Others	350.30	58.9%
	Total Sales	595.00	100.0%

China's imports more than 60%

Sales in US exceed 60%

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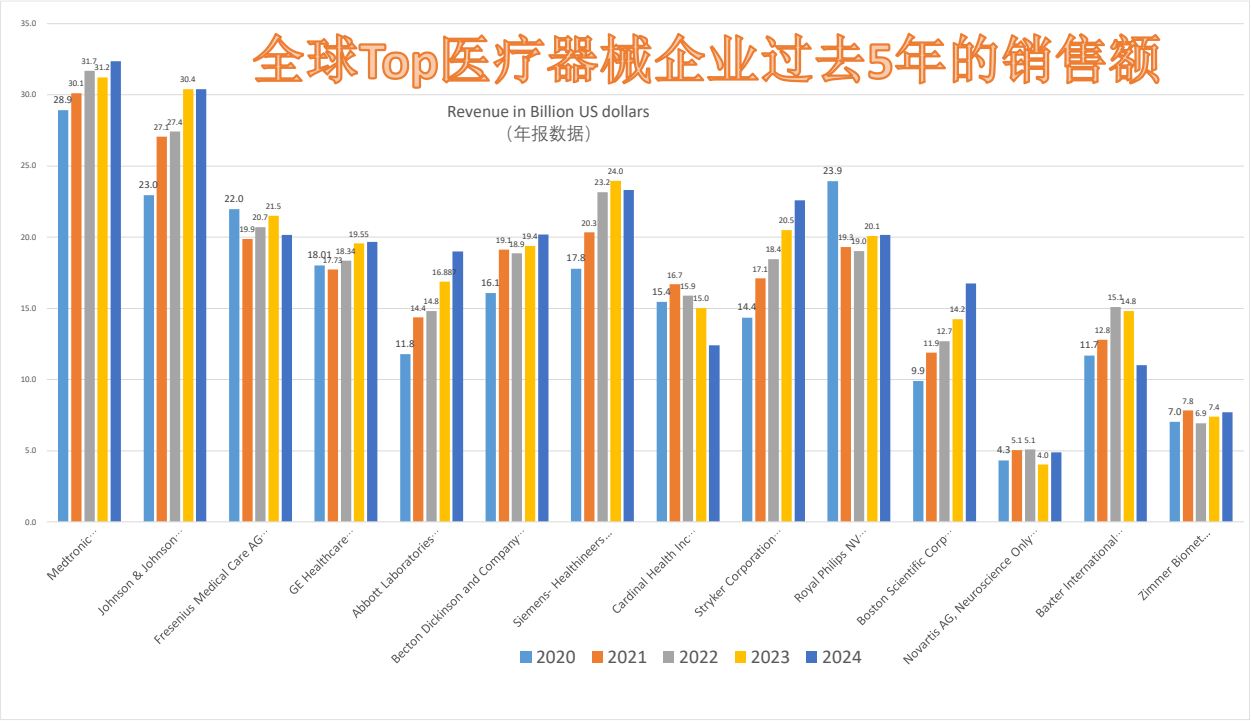
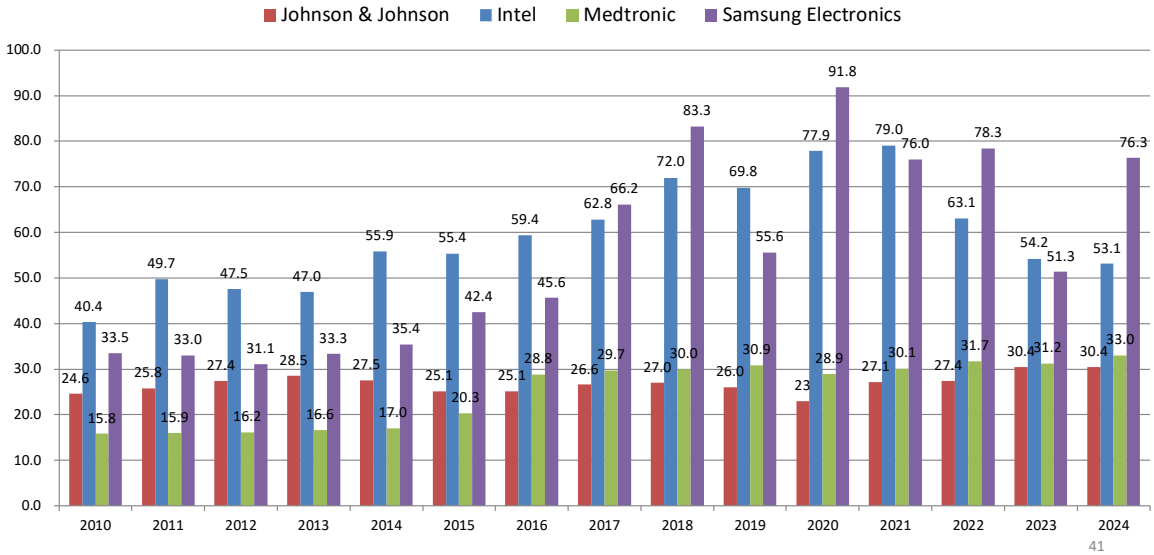


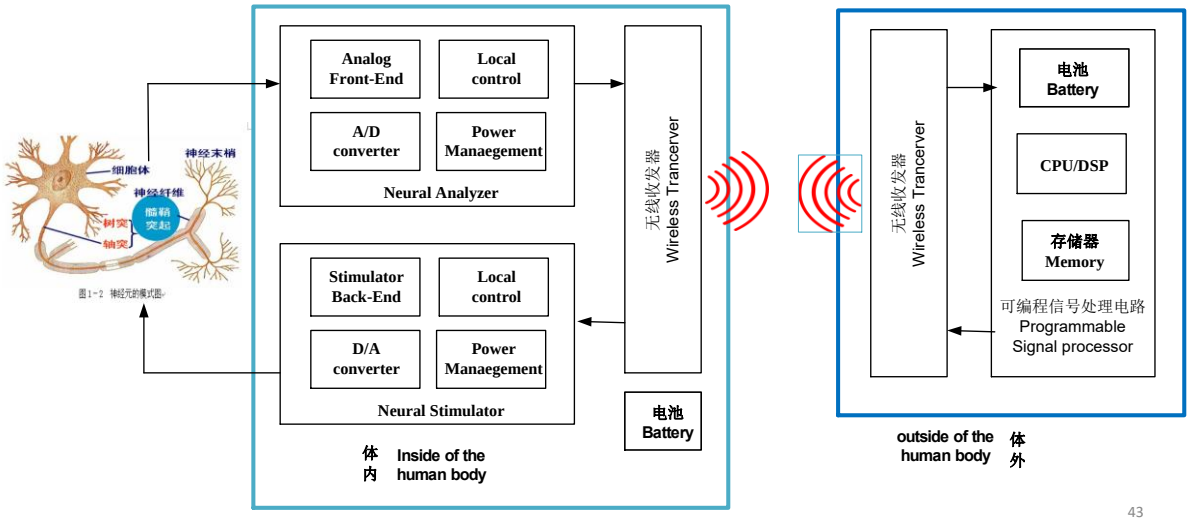
A single enterprise in the medical device industry is smaller than the information industry



实例：神经刺激器及智能助听

Case Studies in Neuroscience and Hearing Aiding

Perspective to neural modulation as electronic technology(closed-loop)



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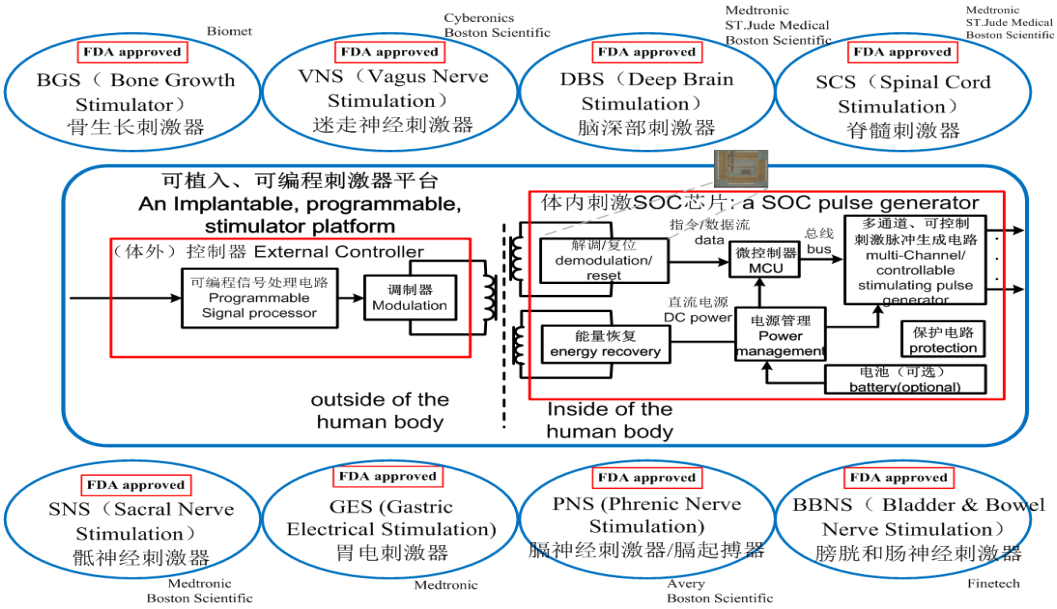
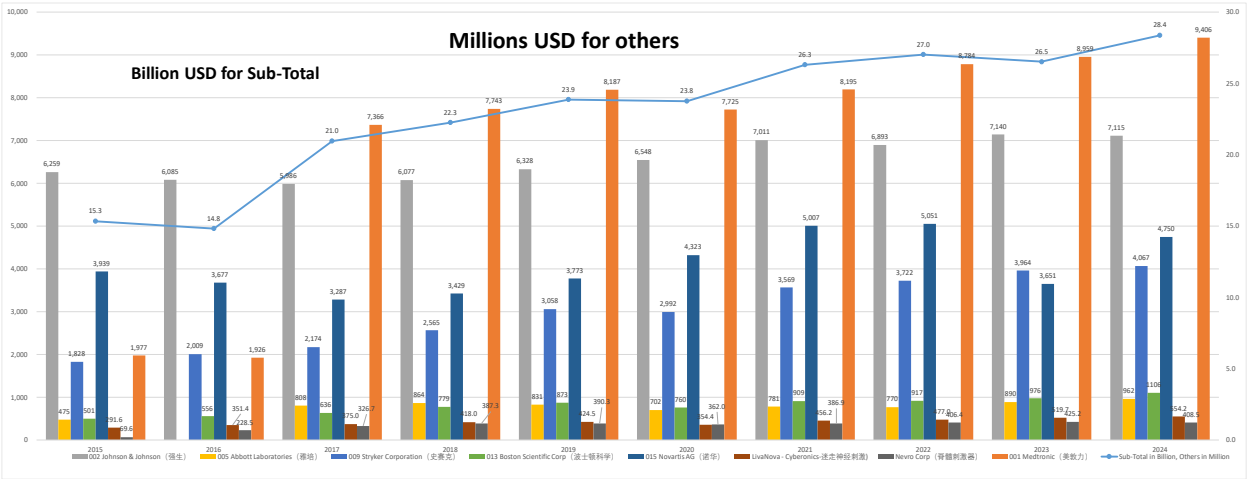
Possible applications of neural modulation



The revenue of global major companies selling neural modulation devices in last decade
全球主要销售神经处理设备的企业近十年的销售额

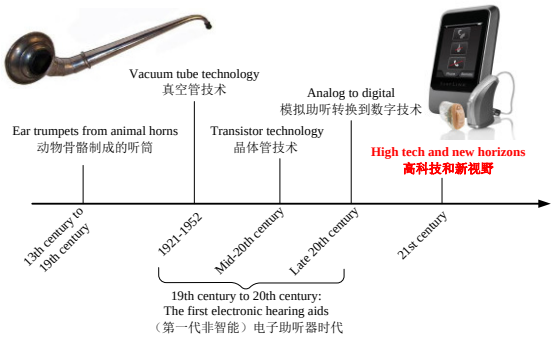
Key point: The market size of neural modulation devices is large enough. Total sales of neural modulation devices by the 8 companies in 2024 are about 28.4 billion US dollars

重点：神经刺激器械的市场规模足够大，图中8个企业2022年神经刺激器械销售总额约284亿美元



Hearing aid history: From ear trumpets to digital technology
助听器简史：从听筒到数字技术

- 13世纪早期，听力损失的人们开始掏空牛或绵羊等动物的角作为原始听力改善设备。 18 世纪，发明了更时髦的听筒，人们第一次试图治疗听力损失，采用了漏斗型设计，尽管不能放大声音，但可以收集声音并采用漏斗让声音通过窄管进耳朵。
- 19世纪，电话开始应用，很快人们发现有些听力损伤的人可以通过电话交流，而面对面反而听不到声音。Thomas Edison对放大声音信号问题做了研究，他将电话中的碳送话器改造成放大器，可以放大15dB。由于助听的应用需要30dB的增益，爱迪生的这个发明尽管没有投入使用，但为电子助听器的发明奠定了基础。20年代开始，真空管的增益可以实现70dB，1924年开始，真空管的尺寸减小到可以装在一个木盒子用于助听，用手持受话器和送话器实现助听。问题：大、重、耗电、放大全部声音而不是只放大患者想听的声音。 1938 Aurex发明了第一个可以佩戴的真空管助听器，其送话器和受话器可以夹在衣服上。但电池包的依然很大、很重，需要绑在用户的腿上。第二次世界大战期间，技术发展很快，1940年代实现了部件装在电路板上、电池尺寸减小，使得完整的助听器的尺寸可以放在一个背包里。



- 1948年，Bell实验室发明晶体管。同年，Raytheon公司的工程师Norman Krim开始用晶体管设计电子助听器，1952年 Krim建立了采用结型场效应管技术的助听器公司，产品畅销，1953年一年就销售20万台助听器。
- 1960s年底， Zenith Radio 公司提出的助听器结构（模拟助听）一直应用到1990年代。2000年之后逐步采用数字化技术，增加可编程功能、允许个性化等等。到2005年，数字助听器的销售量就达到了80%。

http://www.healthyhearing.com/report/47717-Digital-hearing-aid-history46

数字技术引起现代助听器的变化

How Digital Technology Has been Changed Modern Hearing Aids

At that time, the idea of a computer being small enough to fit on or in the ear was viewed as science fiction, and as one wag put it, "It may be a good hearing aid, but you'll need a friend with a wheelbarrow behind you to carry the instrument."

三种输入方式：
麦克风、FM及磁带录音机

阵列处理数字助听器及使用者
（纽约城市大学）
Photograph of the array processor digital hearing aid and a person using the system. City University of New York

信号处理用于助听的基础：离线DSP
Signal Processing for Hearing Aids Applications: Offline DSP

The first hearing aid consisted of a body worn processor with a hardwire connection to ear mounted transducers.

Audible Difference, a company provides audiological care in Madison, Wisconsin

The first BTE digital hearing aid was developed toward the end of the project

具备声音放大功能及复杂信号处理功能的复杂仪器
Today, the hearing aid is a far more complex instrument with advanced forms of signal processing

语音增强，噪声消除，自适应方向选择，反馈声消除，数据监测，声学场景识别及分析，与其他通信系统的无线链接，双耳声音放大
Speech Enhancement, Noise Reduction, Self-adapting Directional Inputs, Feedback Cancellation, Data Monitoring, Acoustic Ccene Analysis, Wireless Link with other communication Systems, Binaural Amplification

数字助听器的前奏：
音频信号的数字化
Prelude to Digital Hearing Aids: Digitizing Audio Signals

计算机控制的
模拟放大器
Computer-Controlled Analog Equipment

实时DSP：
数字助听器
DSP in Real Time: The Digital Master Hearing Aids

单纯放大声音
Amplify sound only

数字助听器的出现，与采样定理的提出密切相关
Relevance of the Sampling Theorem for Modern Hearing Aids

Pre-digital era 模拟时代 | Project X, was initiated in the early 1940s and closely guarded secret until 1975 | 1948 | Mid-1960s | late 1970s | late 1984 - 1989 | Today (2017) 现在

The Nicolet Corporation is credited with developing the first commercial digital hearing aid using custom fabricated Chips. Nicolet's Project Phoenix under the direction was initiated in 1984 and ended in 1989, when Nicolet withdrew from the hearing aid market.

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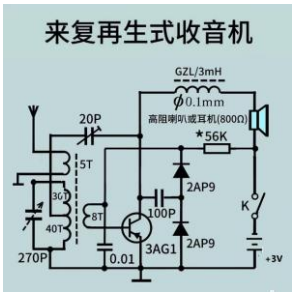
集成电路：信息时代的核心（晶体管阶段）



信息产品（应用）

1952-1953：助听器（Hearing Aids），Sonotone Corporation 和 Zenith 等公司

1954：便携式收音机（Portable Radios），Regency（与德州仪器合作）

整个1950年代，军用与航天设设备。美国军方看到了晶体管的巨大潜力。其坚固、可靠、省电的特性对便携式电台、声纳系统、导弹至关重要。军方的大量订单是驱动早期晶体管技术发展和产能扩张的主要力量。

1950年代中后期，1955年，IBM推出了IBM 608，这是第一款完全使用晶体管而非真空管的商用计算机。晶体管计算机更小、更快、更可靠，标志着计算机"第二代"的到来。

集成电路（晶体管）

1947：贝尔实验室发明晶体管

1951：首批商用晶体管在Western Electric产生

1951：贝尔实验室举办了著名的"晶体管技术研讨会"，向全球35家公司（包括GE、RCA、Philco、德州仪器等）授权了晶体管技术，费用仅为2.5万美元

1952：结型晶体管的诞生，Western Electric和Raytheon 等企业生产。

1954年：硅晶体管的诞生，Texas Instruments，通常被认为是晶体管革命的引爆点

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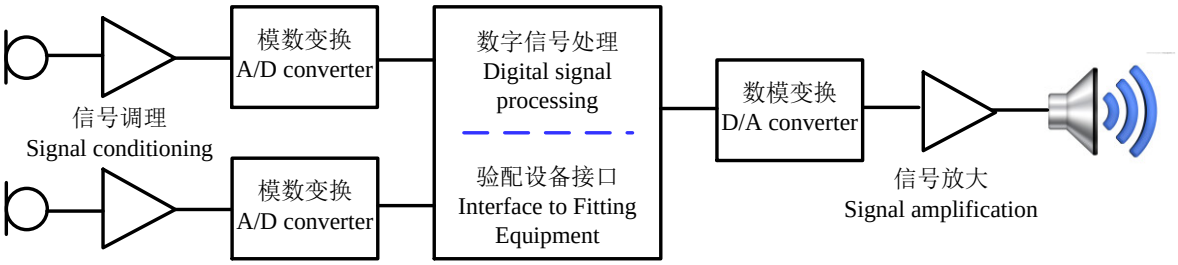
Perspective of hearing aid as electronic technology (open-loop)

Composition of a traditional single ear hearing aid with double microphones

传统单耳双麦克风助听器组成



49

Complex in fitting of hearing aid

Expensive in price of Hearing aids

Consumption habits

Global hearing aids market during the period 2016–2024

- The global hearing aids market size was valued at USD 10.1 billion in 2021 and is expected to expand at a compound annual growth rate (CAGR) of 4.9% from 2022 to 2030.

- 2021全球助听器市场规模为101亿美元（其中美国36.6亿美元），预计2022年至2030年将以4.9%的复合年增长率（CAGR）扩张。

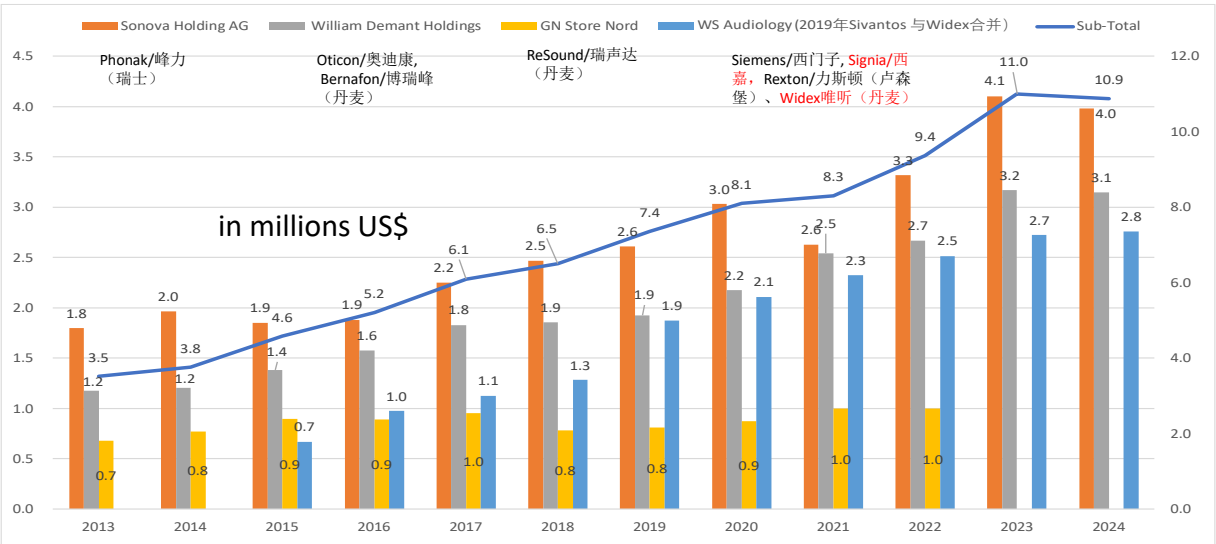
  - https://www.grandviewresearch.com/industry-analysis/hearing-aids-market

真有这么大？！

Is it really so large?!

51

Hearing Aids Market Size was valued at USD 8.39 Billion in 2018 and is anticipated to reach USD 14.45 Billion by the end of 2026. The forecast period is set from 2019 to 2026 and the market is expected to rise at a CAGR of 7.2%.



Sales of the top hearing aid companies in last 10 years

应用于医疗仪器的集成电路

如何选题

53

It should be a device with a Medical-grade

- An instrument, apparatus, implement, machine, contrivance, implant, in vitro reagent, or other similar or related article, including a component part, or accessory which is:
  - Recognized in the official National Formulary
  - Intended for use in the diagnosis of disease or other conditions
  - Intended to affect the structure or any function of the body of man or other animals

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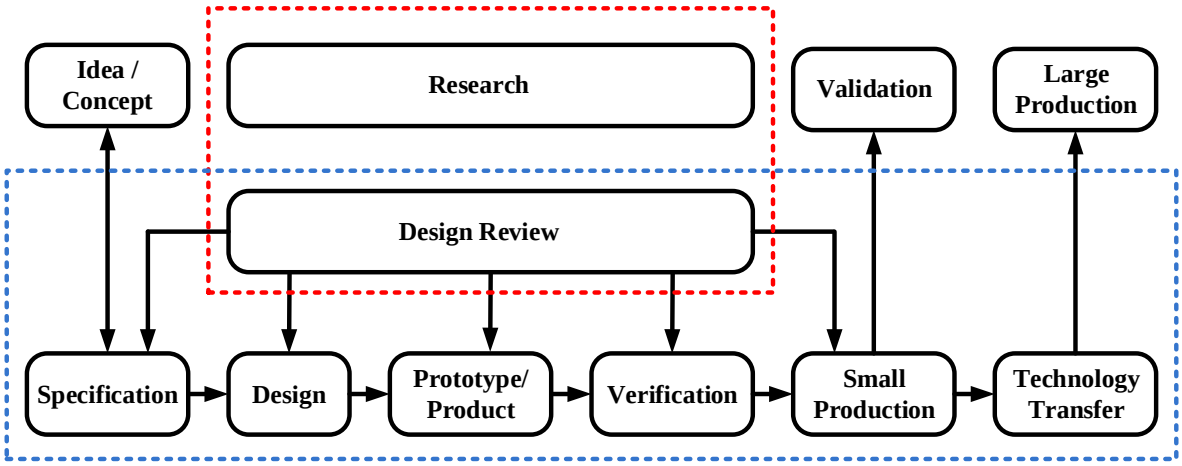
Classification of medical Devices

- Class I: General controls
- Class II: General controls with special controls
  - infusion pumps, and surgical drapes…
- Class III: General controls and premarket approval
  - implantable pacemaker, pulse generators, automated external defibrillators…

55

Research and Development of an medical system



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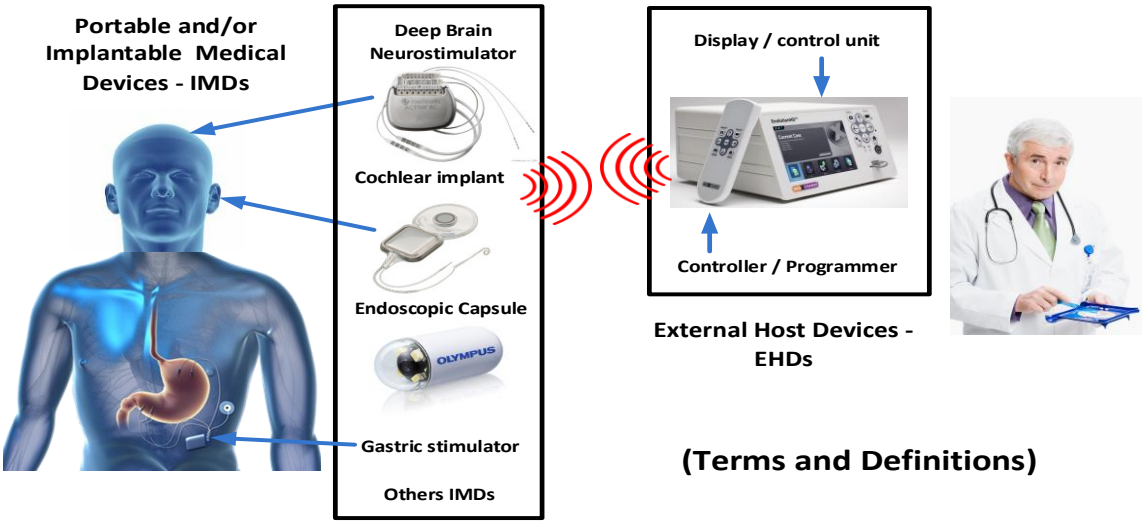
A Potable and/or Implantable Device

- Electronic implantable medical devices (IMD) are designed to be fully or partially implanted in the human bodies through surgeries[1], and remain in bodies for several hours to several years or even permanently after the surgical intervention.

What is a Portable and/or Implantable Medical Device?

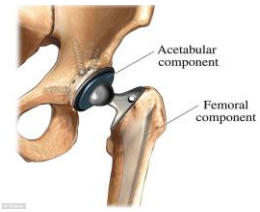
[1] R. Ritter, J. Handwerker, T. Liu, and M. Ortmanns, "Telemetry for Implantable Medical Devices," IEEE SOLID-STATE CIRCUITS MAGAZINE, vol. 6, Issue. 2, pp. 47-51, Spring 2014.

A Medical System with Portable and/or Implantable Medical Devices

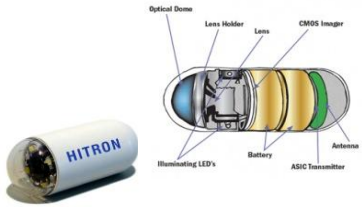


(Terms and Definitions)

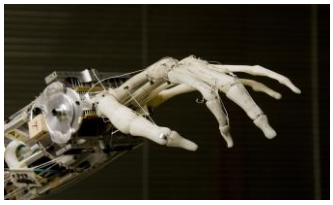
Examples of Potable and/or Implantable Medical Devices



Total hip replacement



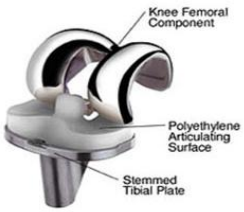
capsule endoscopy



Robot hand

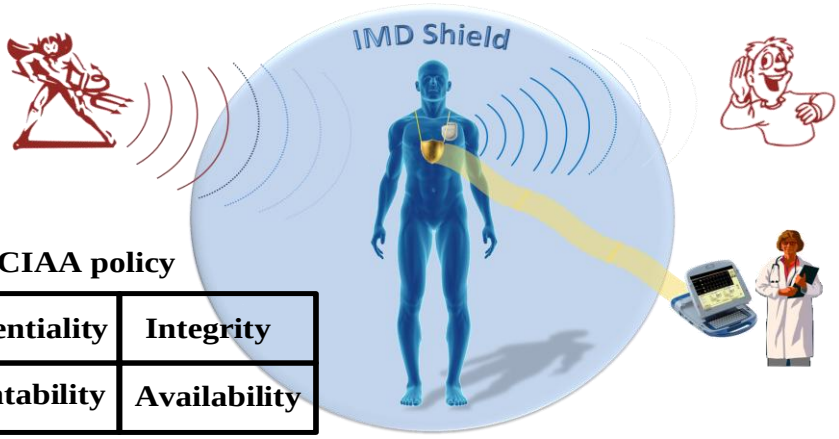
Cochlear implants

Nerve Stimulator



Total Knee replacement

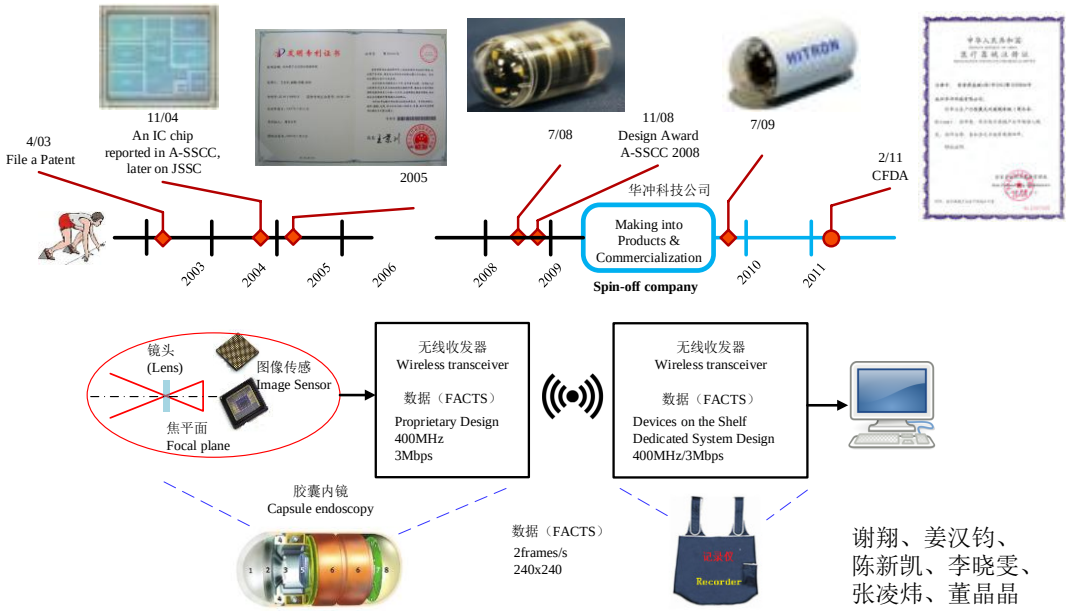
About the information security - Mostly at the system level and implemented in software



CIAA policy

Confidentiality	**Integrity**
Accountability	**Availability**

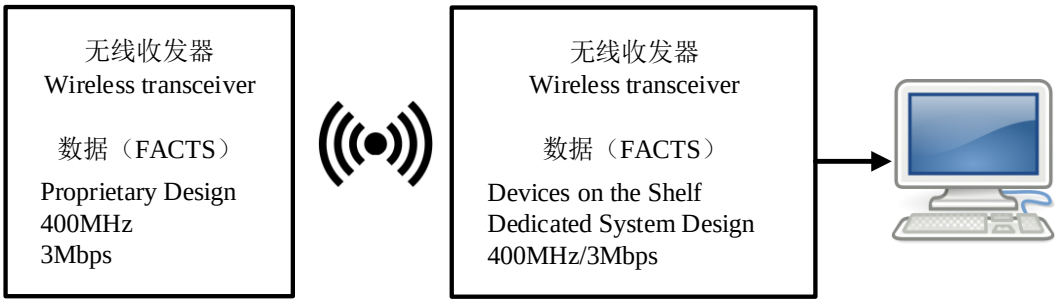
口服胶囊内视镜及其电路设计



谢翔、姜汉钧、
陈新凯、李晓雯、
张凌炜、董晶晶

61

假定有了这个收发器组，我们还能做什么？



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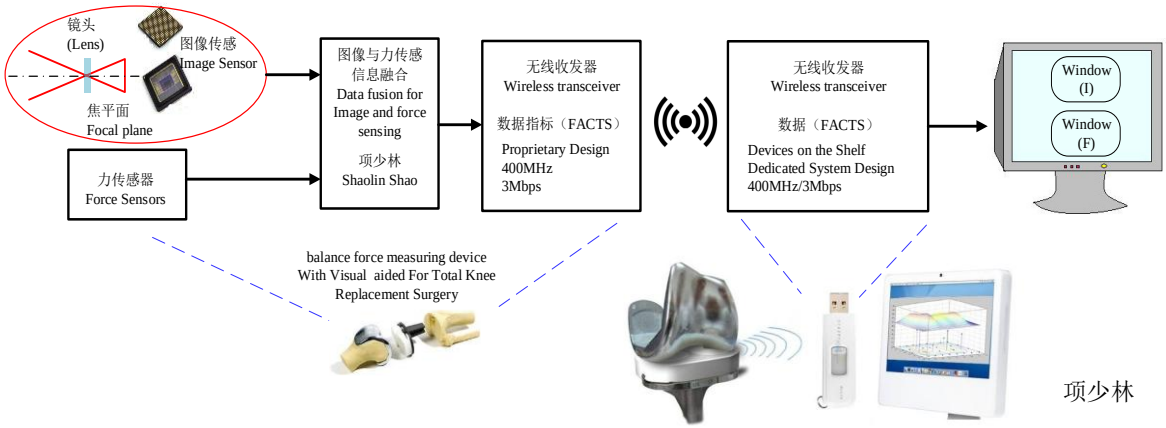
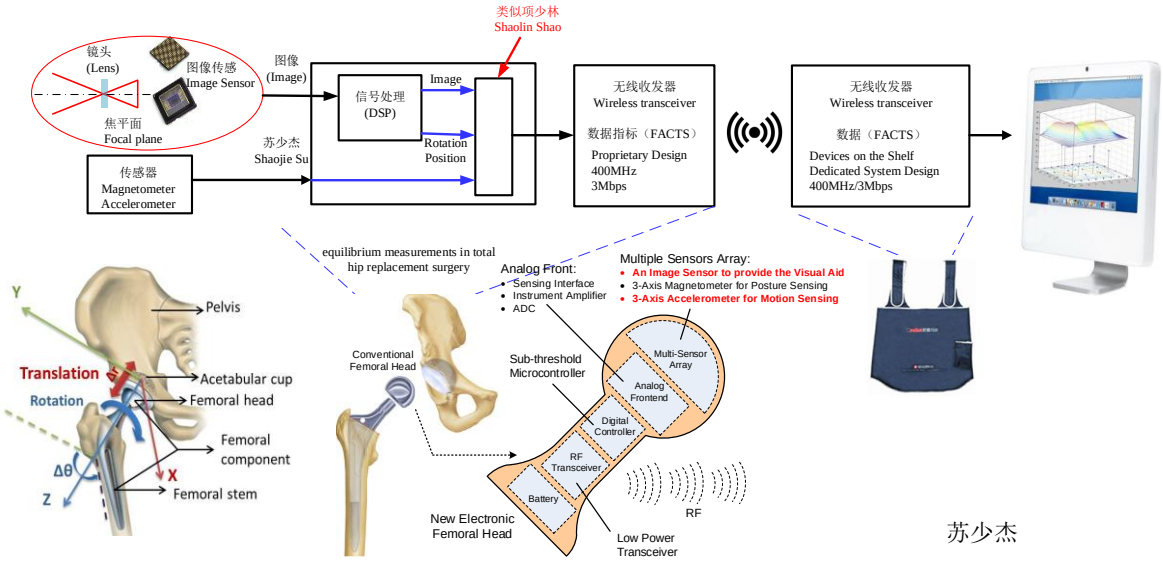
全膝关节置换手术平衡测量器



项少林

63

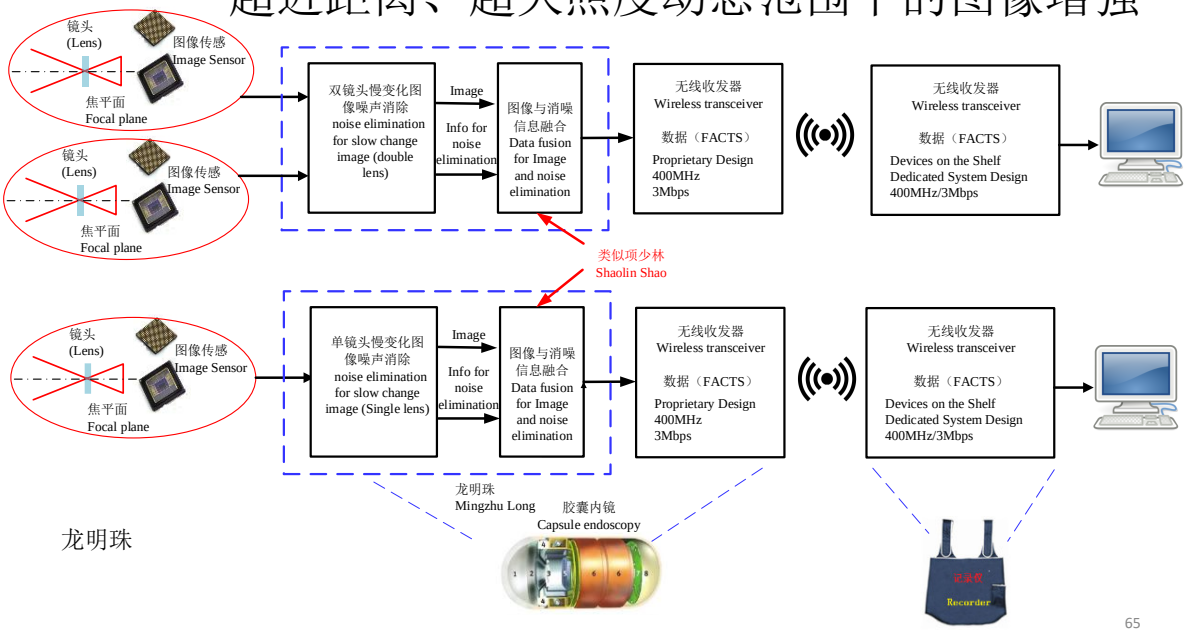
髋关节置换手术平衡测量器



苏少杰

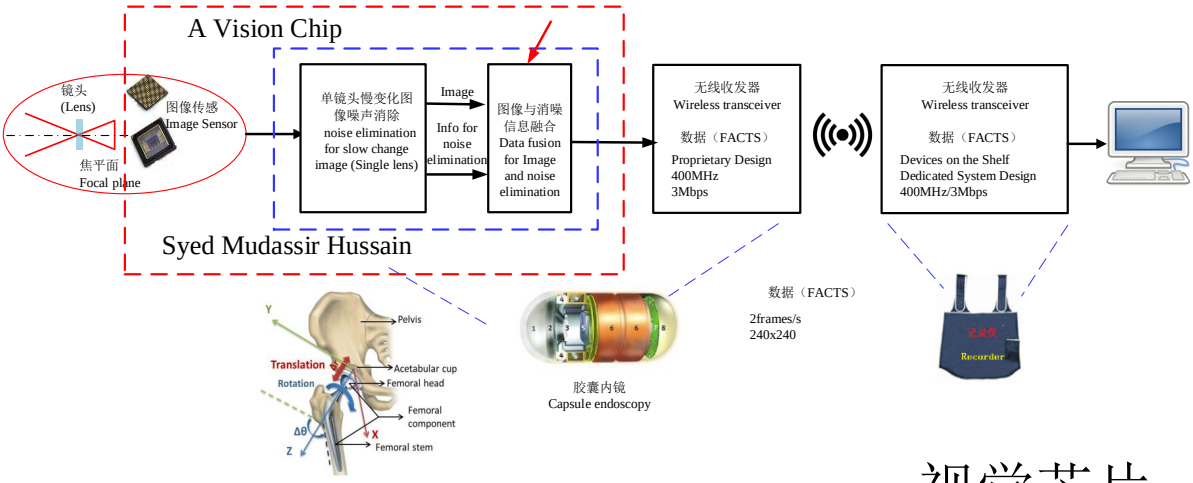
64

超近距离、超大照度动态范围下的图像增强



龙明珠

65



视觉芯片
(A Vision Chip with artificial intelligence)

66

医疗电子近期成果

神经刺激器

- 2014年9月获得III类医疗器械生产许可证
- 2020年1月获得III类医疗器械注册证（迷走神经刺激器）
- 拥有10万级洁净厂房和检测、生产、研发等各种设备
- 植入式脊髓刺激器2019年12月完成第一例临床植入
- 临床医院：宣武医院，上海仁济医院，306医院，304医院，北京友谊医院

中华人民共和国医疗器械注册证

注册人名称	常州瑞华[illegible]医疗器械有限公司
注册人住所	常州市新北区[illegible]号213000-D208
生产地址	常州市新北区[illegible]号213000-D208
代理人名称	/
代理人住所	/
产品名称	植入式迷走神经刺激器组件
型号、规格	型号：VNS201，规格A
结构及组成	产品由迷走神经刺激器、扭力扳手、电刺棒和螺旋手术钻组。
适用范围	与特定的植入式迷走神经刺激器电极导电组件配套，供医生根据需要在术中（用于术前、术后及以上）使用，用于癫痫辅助发作频率的缓解治疗。
附件	产品技术要求
其他内容	/
备注	植入物使用期限：长期使用。临床试验中已治疗病人植入物使用期已经超过[illegible]年。[illegible]。

审批部门：国家药品监督管理局

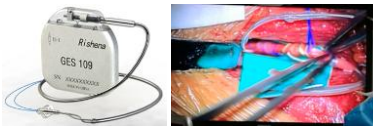
批准日期：[illegible]年[illegible]月[illegible]日

有效期至：[illegible]年[illegible]月[illegible]日

植入式迷走神经刺激器

国械注准20203120085

植入式胃电刺激器



植入式脊髓刺激器

常州瑞神安医疗器械有限公司

- Rishena Medical Co., Ltd. .

https://www.rishena.com/

- An innovative enterprise focusing on the research, development, production and sales of active implantable medical devices

动态体音记录仪

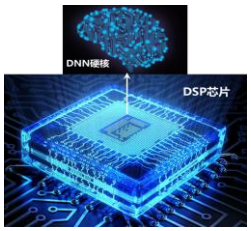
- 临床肠鸣音连续监测、全球首创
- 2018年获得II类医疗器械产品注册登记证、医疗器械生产许可证
- 已在全国多家医院推广使用

鲁械注准20182210271

医保局新增服务项

智能双耳助听器

- 与深圳市龙岗区耳鼻喉医院合作，已获得10000+听损患者数据，完成科研级临床听力测试与补偿验证36人次
- 2017年获中国深圳创新创业大赛行业决赛二等奖、龙岗区第一名
- 已通过上海市医疗器械检验所检验
- 2020年获得医疗器械生产质量体系考核通过证明
- 2020年12月获得II类医疗器械注册证



- ✓ 隐蔽美观
- ✓ 无堵耳效应
- ✓ 轻松整日佩戴

- ✓ 双耳助听
- ✓ 无线验配
- ✓ 隐蔽舒适

- ✓ AI增强的多核DSP芯片
- ✓ 深度学习语音增强和声音场景分类

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深圳市智听科技有限公司（挚听助听器）

- Shenzhen Eartech Company Limited.

https://www.eartech.cn/

- Eartech is the pioneer of smart hearing aids, led by renowned experts and professors from Tsinghua University.

The team is dedicated to the development of intelligent hearing testing and assistant systems, benefiting hundreds of millions of people with hearing loss.

Product Concept

Dr. Fei Chen, based on years of research at the Tsinghua Hearing Aid Lab, proposed the concept of an intelligent hearing aid capable of self-fitting. He has published multiple papers when he was in Tsinghua University.

2016

Prototype

2017

Formed the vision of creating a smart hearing aid that is clear and affordable for everyone.
The company was established to start product development and constructed an independent R&D and production base.

Medical Qualification

Eartech obtained the Class II Medical Device Registration Certificate and Medical Device Manufacturing License.

2020

Release RIC 1.0 / RIC 1.0 Max

2021

Eartech launched the world's first RIC-style smart hearing aid that allows users to conduct the fitting at home independently.
The product was launched on Xiaomi's crowdfunding platform, raising over 5 million yuan.

Release RIC 1.0 Pro

2022

Eartech launched a rechargeable RIC-style hearing aid that supports Bluetooth streaming connectivity. Collaborated with Tencent Ethereal Lab to develop the Tencent Ethereal × Tsinghua Hearing Aid Algorithm

Release ITE 1.0 / ITE 1.0 Pro

2023

Eartech launched a fashionable, TWS-like ITE-style hearing aid.
For patients with tinnitus, we introduced a warble-tone hearing test and tinnitus-relief features.
Sound-field detection FED ensures scientific and effective audiometry.

Comprehensive Upgrade: Launch of RIC 2.0 Series

2024

【RIC 2.0 Pro】
- A major new product release with comprehensive upgrades.
- Audiometry and fitting via WeChat Mini Program.
- Face-to-face video remote tuning services via Tencent Meeting.
- Tencent Ethereal x Tsinghua Hearing Aid Algorithm 2.0.

【RIC 2.0 Pro Max】
- High-power Hi-Fi balanced armature driver.
- Deep noise reduction algorithm.
- Clinical validation by Beijing Tongren Hospital.

膝关节置换智能辅助系统

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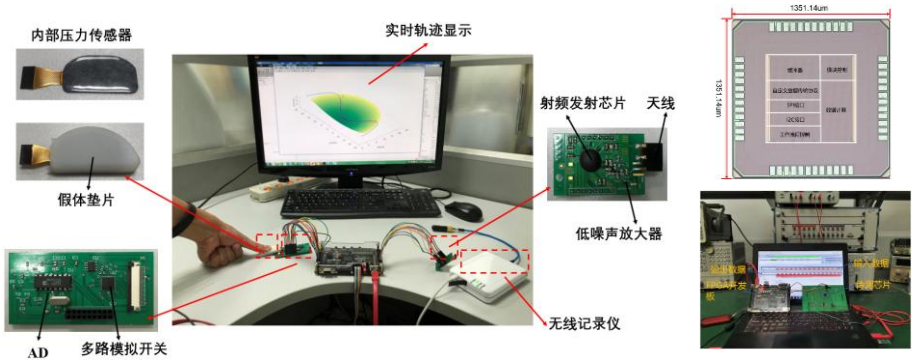
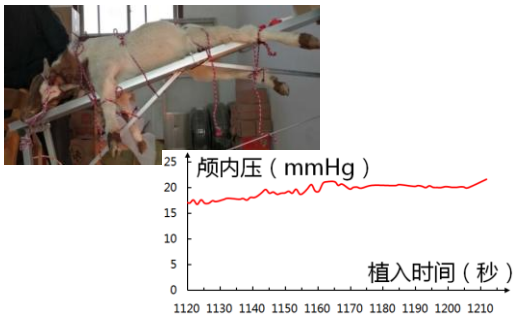
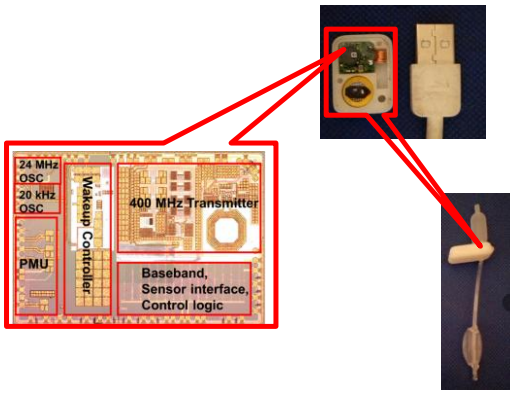
膝关节置换智能辅助系统

- **最新进展：首次提出一种单髁膝关节置换压力及力线轨迹智能检测系统**
  - 首次采用集成电路技术以及传感器技术设计实现了单髁膝关节置换压力及力线轨迹智能检测系统；
  - 解决了力线轨迹实时拟合问题以及最大压力值点的检测问题、提出了力线轨迹拟合算法等。
  - 研究成果属国内外首创。



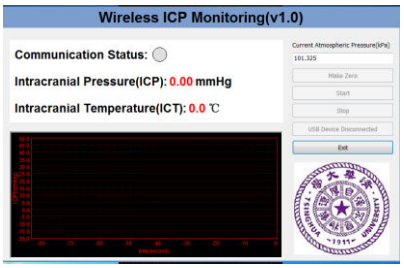
颅内压监测





Implantable Device
22x17x8 mm^3

Wireless Data Recorder



Computer Software

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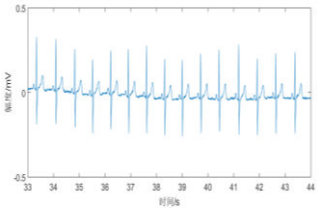
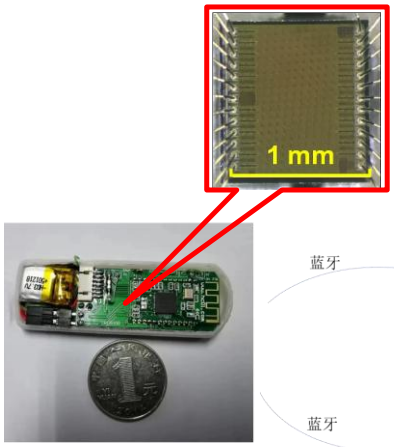
植入式心电监护

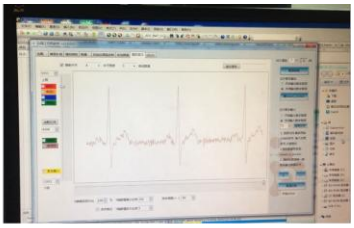




植入式心电监护仪

蓝牙

蓝牙



PC心电监护软件

结束语

“求其上者得其中，求其中者得其下，求其下者……”。

唐太宗《帝范》卷四：“取法于上，仅得为中，取法于中，故为其下。”这句话的意思是，追求上等的，可以得到中等的，追求中等的，只能得到下等的。

宋末元初时期的诗词评论家严羽在其《沧浪诗话》中曰：“学其上，仅得其中；学其中，斯为下矣。”向上等的学习，只能达到中等水平；向中等的学习，只能达到下等水平。指学习的目标必须定得高。

调研内容（作业）

- 选择一
  1. 粗读最近数期IEEE Trans. on Biomedical Engineering
  2. 延续粗读最近的EMBS(IEEE Engineering in Medicine and Biology Society ）的旗舰会议EMBC: International Engineering in Medicine and Biology Conference(EMBC)的会议录
  3. 选择某种适合用电信号处理的疾病，了解“观测、处理” 这种疾病的最新学术进展
  4. 精读数篇用电信号观测、处理这种疾病的学术论文，写出调研报告
- 选择二

  粗读最近数期IEEE Transactions on Biomedical Circuits and Systems；延续粗读最近的CASS(IEEE Circuit and Systems Society ）的旗舰会议The International Symposium on Circuits and Systems (ISCAS)以及Biomedical Circuits and Systems Conference （BioCASS）会议录；选择某种电路，适合处理某种疾病疾病，这种电路研究的最新学术进展；精读数篇这种电路的学术论文，写出调研报告
- 选择三
  - 研究某种医疗器械，其中采用的核心技术是电子技术，了解不同医疗器械企业在这种医疗器械中采用的电子技术，研究其器械研发的可能性
- 要求：

  阅读文献量不少于8篇，精读不少于4篇，报告结尾附参考文献，报告正文字数不少于5000字）。